

A PROJECT Title
ON
“BLOOD BANK MANAGEMENT SYSTEM”
Insem-End project submitted in partial fulfilment of the requirements for the Course
Front End Development Frameworks for the degree of
BACHELOR OF TECHNOLOGY
IN
Computer Science and Engineering
(2025-2026 A.Y)

BY

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(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: **BLOOD BANK MANAGEMENT SYSTEM**

1. Abstract:

Abstract – BLOOD BANK MANAGEMENT SYSTEM

The **Blood Bank Management System** is a web-based application designed to efficiently manage and streamline blood bank operations. The system focuses on maintaining accurate records of blood inventory based on blood groups and components, managing donor information, and ensuring timely availability of blood for patients in need. It provides a centralized platform for handling blood donation, storage, and distribution processes.

The application enables users to register and maintain donor profiles, track donation history, and perform compatibility checks between donors and recipients to ensure safe transfusions. It also includes features for managing blood issue requests, monitoring stock levels, and generating alerts for blood units nearing expiration. This helps reduce wastage and ensures optimal utilization of available resources.

An admin panel is incorporated to oversee operations, manage data, and handle notifications. The system supports data export and integrates REST APIs for smooth communication between frontend and backend. Tools such as React.js, Node.js, Express.js, and optional MySQL database are used to build a responsive and scalable application, while Axios and Postman facilitate API handling and testing.

Overall, the Blood Bank Management System improves efficiency, reduces manual errors, and enhances the reliability of blood bank services through automation and real-time data management.

2. INTRODUCTION:

3. SOFTWARE REQUIREMENTS:

Course Coordinator

Course Instructor

FED , Coordinator